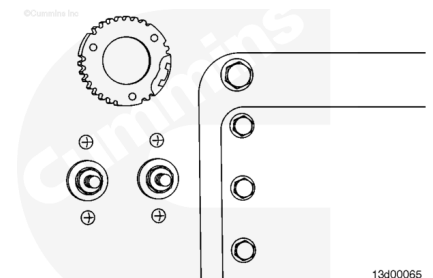


## Trainings ▾

### General Information

In addition to the heavy duty battery connections used for the starter, marine engines equipped with an Electronic Control Module (ECM) and the SmartCraft system use an additional unswitched power supply. The connection is installed directly from the battery to lug style connectors mounted on the side of the engine.

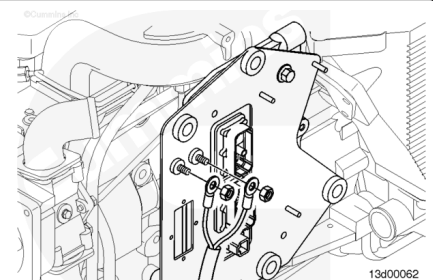
These terminal lug blocks are located on the left side of the engine near the OEM interface connector and the service tool datalink connection. The positive terminal block is red and the negative terminal block is black.



### Initial Check

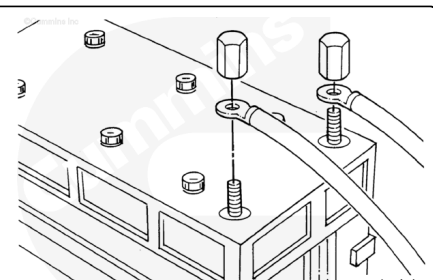
#### **⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**



The unswitched power supply wires connect from the battery to the positive and negative terminal blocks. These wires are supplied by the OEM.

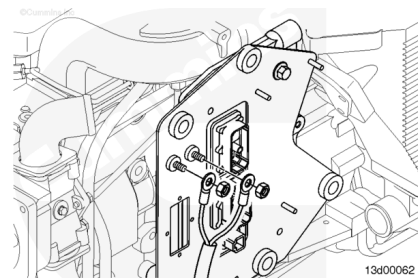
Before removing the unswitched power supply wires from the engine, be sure the positive and negative wires are disconnected from the battery. Refer to procedure 013-009 (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>).



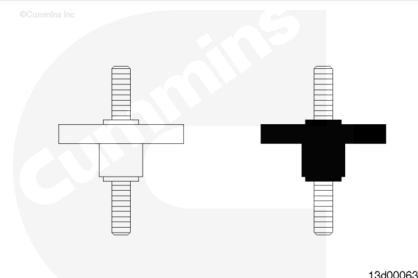
Always install the unswitched positive and negative wire connectors to the terminal blocks before installing the connections to the battery.

Torque the connections to the terminal blocks to the specified value.

**Torque Value:** 5 n·m [ 45 in-lb ]



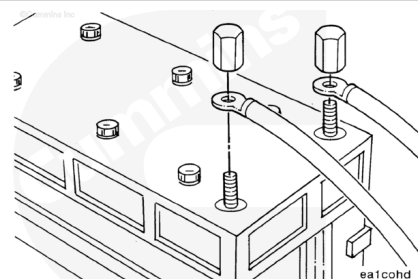
If for some reason the terminal blocks are damaged, new terminal blocks can be obtained and installed.



## Preparatory Steps

### **⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**



Before removing the unswitched power supply wires from the engine, be sure the positive and negative wires are disconnected from the battery. Refer to Procedure 013-009 (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>).

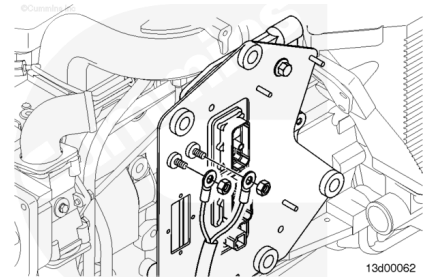
If a battery switch is being used on the vessel, be sure to disconnect all batteries.

Remove the external unswitched power supply battery connection from the battery.

Remove the unswitched wire connectors from the positive and negative terminal blocks.

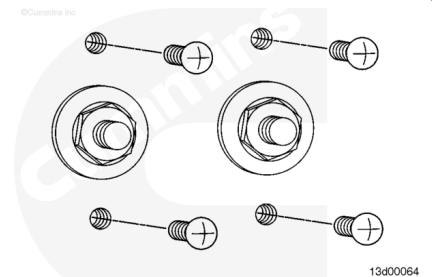
## Remove

Remove the engine wiring harness connectors from the back of the positive or negative terminals. Some plates may have to be removed to complete this operation.



Remove the upper and lower mounting capscrews from the positive or negative terminal blocks. Slide the terminal blocks out the back of the mounting plate.

Some plates may have to be removed to complete this operation.

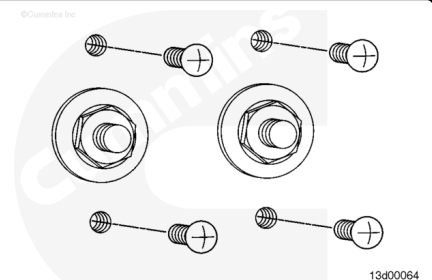


## Install

Install the new positive or negative terminal block through the rear of the mounting plate. On some models the plate will have to be removed to complete this operation.

Install the mounting capscrews and torque to the specified value.

**Torque Value:** 3.5 n·m [ 30 in-lb ]

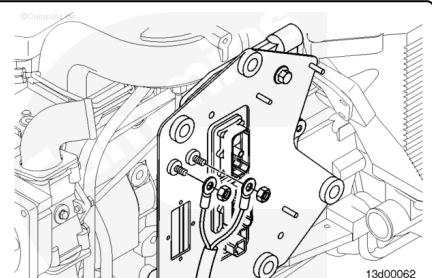


Install the engine wiring harness positive or negative wire connectors to the correct positive and negative terminal blocks located on the back of the mounting plate.

Some plates may have to be removed to complete this operation.

Torques to the specified value.

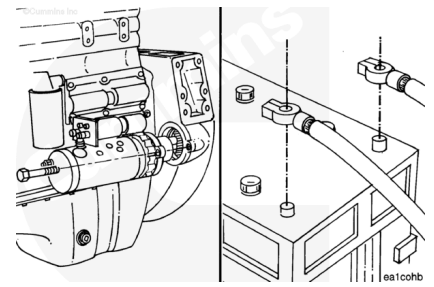
**Torque Value:** 5 n·m [ 45 in-lb ]



## Finishing Steps

### **⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**



Install the unswitched power supply connectors to the positive and negative terminal blocks.

Install the unswitched power supply connectors to the battery and tighten the battery connection.

Ensure the positive and negative wires are connected to the proper battery terminals. Refer to Procedure 013-009 (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>).